

Index.html

```
<!DOCTYPE html>
<html><head><title>Login</title>
<style>
body { font-family: Arial, sans-serif; margin: 60px; background-color: #f4f6f8; }
.box { background: #fff; padding: 25px 35px; border-radius: 8px;
box-shadow: 0 0 10px rgba(0,0,0,0.15); max-width: 400px; margin: auto; text-align:center;}
input[type="text"], input[type="password"] { padding: 6px; width: 80%; margin: 10px 0; }
input[type="submit"] { background-color: #2c3e50; color: #fff; border: none;
padding: 10px 20px; border-radius: 5px; cursor: pointer; }
</style></head>
<body>
<div class="box">
<h2>Login</h2>
<form action="WelcomeServlet" method="post">
Username: <input type="text" name="username"><br>
Password: <input type="password" name="password"><br>
<input type="submit" value="Login">
</form>
</div>
</body>
</html>
```

WelcomeServlet.java

```
import java.io.IOException; import java.io.PrintWriter; import javax.servlet.ServletException; import
javax.servlet.annotation.WebServlet; import javax.servlet.http.Cookie; import javax.servlet.http.HttpServlet; import
javax.servlet.http.HttpServletRequest; import javax.servlet.http.HttpServletResponse; import
javax.servlet.http.HttpSession;
@WebServlet("/WelcomeServlet") public class WelcomeServlet extends HttpServlet {
protected void doPost(HttpServletRequest request, HttpServletResponse response)
throws ServletException, IOException {
String username = request.getParameter("username");
String password = request.getParameter("password");
response.setContentType("text/html");
PrintWriter out = response.getWriter();
if (username == null || username.trim().isEmpty() || password == null || password.trim().isEmpty()) {
out.println("<!DOCTYPE html><html><head><title>Login Error</title>");
out.println("<style>body{font-family:Arial;margin:60px;background:#f4f6f8;text-align:center;}</style>");
out.println("</head><body>");
out.println("<h2 style='color:red;'>Username or Password cannot be empty!</h2>");
out.println("<a href='index.html'>Go Back</a>");
out.println("</body></html>");
return;
}
String passwordRegex = "^(?=.*[a-z])(?=.*[A-Z])(?=.*[0-9])(?=.*[!@#%&*^a-zA-Z0-9]).+$";
if (password == null || !password.matches(passwordRegex)) {
out.println("<!DOCTYPE html><html><head><title>Password Policy Error</title>");
out.println("<style>body{font-family:Arial;margin:60px;background:#f4f6f8;text-align:center;}</style>");
out.println("</head><body>");
out.println("<h2 style='color:red;'>Password does not meet security requirements!</h2>");
out.println("<p>Password must include at least one lowercase letter, one uppercase letter, one number, and one
special character.</p><br>");
out.println("<a href='index.html'>Try Again</a>");
out.println("</body></html>");
return;
}
HttpSession session = request.getSession();
session.setAttribute("user", username);
Cookie cookie = new Cookie("userCookie", username);
response.addCookie(cookie);
synchronized (getServletContext()) {
Integer visitorCount = (Integer) getServletContext().getAttribute("visitorCount");
if (visitorCount == null) {
visitorCount = 0;
}
if (session.getAttribute("counted") == null) {
visitorCount++;
getServletContext().setAttribute("visitorCount", visitorCount);
session.setAttribute("counted", true);
}
}
Integer visitorCount = (Integer) getServletContext().getAttribute("visitorCount");
out.println("<!DOCTYPE html><html><head><title>Welcome</title>");
```

```

        out.println("<style>body{font-family:Arial;margin:60px;background:#f4f6f8;}");
        out.println(".box{background:#fff;padding:25px 35px;border-radius:8px;}");
        out.println("box-shadow:0 0 10px rgba(0,0,0,0.15);max-width:450px;margin:auto;}");
        out.println("a{display:block;margin:10px 0;color:#2c3e50;font-weight:bold;text-decoration:none;}");
        out.println("a:hover{text-decoration:underline;}");
        out.println("</style></head><body>");
        out.println("<div class='box'>");
        out.println("<h2>Welcome, " + username + "</h2>");
        out.println("<p><b>Total Unique Visitors So Far: " + visitorCount + "</b></p>");
        out.println("<h3>1. Hidden Form Field Demo</h3>");
        out.println("<form action='HiddenFieldServlet' method='post'>");
        out.println("<input type='hidden' name='hf' value='" + username + "'>");
        out.println("<input type='submit' value='Go (Hidden Field)'>");
        out.println("</form>");
        out.println("<h3>2. URL Rewriting Demo</h3>");
        out.println("<a href='URLRewriteServlet?username='" + username + "'>Visit (URL Rewriting)</a>");
        out.println("<h3>3. Cookie Demo</h3>");
        out.println("<a href='CookieServlet'>Visit (Cookies)</a>");
        out.println("<h3>4. HttpSession Demo</h3>");
        out.println("<a href='SessionServlet'>Visit (HttpSession)</a>");
        out.println("</div></body></html>");
    }
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
        response.getWriter().println("Please login on the home page first.");
    }
}

```

VisitorCountServlet.java

```

import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletContext;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.HttpSession;
@WebServlet(urlPatterns = {"/VisitorCountServlet"})
public class VisitorCountServlet extends HttpServlet {
    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
        HttpSession session = request.getSession();
        ServletContext context = getServletContext();
        synchronized (context) {
            Integer count = (Integer) context.getAttribute("visitorCount");
            if (count == null) count = 0;

            if (session.getAttribute("visited") == null) {
                count++;
                context.setAttribute("visitorCount", count);
                session.setAttribute("visited", true);
            }
        }
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        out.println("Unique Visitors: " + context.getAttribute("visitorCount"));
    }
}

```

URLRewriteServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
@WebServlet("/URLRewriteServlet")
public class URLRewriteServlet extends HttpServlet {
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
        String username = request.getParameter("username");
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        out.println("<!DOCTYPE html><html><head><title>URL Rewriting Result</title>");
        out.println("<style>body{font-family:Arial;margin:60px;background:#f4f6f8;}");
        out.println(".box{background:#fff;padding:25px 35px;border-radius:8px;}");
        out.println("box-shadow:0 0 10px rgba(0,0,0,0.15);max-width:400px;margin:auto;text-align:center;}");
        out.println("</style></head><body>");
        out.println("<div class='box'>");
        out.println("<h2>Hello, " + username + "</h2>");
        out.println("<p>(Passed via URL Rewriting query parameter.)</p>");
        out.println("</div></body></html>");
    }
}
```

SessionServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.HttpSession;
@WebServlet("/SessionServlet")
public class SessionServlet extends HttpServlet {
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
        HttpSession session = request.getSession(false);
        String name = (session != null) ? (String) session.getAttribute("user") : "Guest";

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        out.println("<!DOCTYPE html><html><head><title>HttpSession Result</title>");
        out.println("<style>body{font-family:Arial;margin:60px;background:#f4f6f8;}");
        out.println(".box{background:#fff;padding:25px 35px;border-radius:8px;}");
        out.println("box-shadow:0 0 10px rgba(0,0,0,0.15);max-width:400px;margin:auto;text-align:center;}");
        out.println("</style></head><body>");
        out.println("<div class='box'>");
        out.println("<h2>Hello, " + name + "</h2>");
        out.println("<p>(This value was retrieved securely via server-side HttpSession storage.)</p>");
        out.println("</div></body></html>");
    }
}
```

HiddenFieldServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
@WebServlet("/HiddenFieldServlet")
public class HiddenFieldServlet extends HttpServlet {
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
        String name = request.getParameter("hf");
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        out.println("<!DOCTYPE html><html><head><title>Hidden Field Result</title>");
        out.println("<style>body{font-family:Arial;margin:60px;background:#f4f6f8;}");
        out.println(".box{background:#fff;padding:25px 35px;border-radius:8px;}");
        out.println("box-shadow:0 0 10px rgba(0,0,0,0.15);max-width:400px;margin:auto;text-align:center;}");
        out.println("</style></head><body>");
        out.println("<div class='box'>");
        out.println("<h2>Hello, " + name + "</h2>");
        out.println("<p>(This value was passed via a Hidden Form Field, ");
        out.println("invisible to the user but submitted along with the form.)</p>");
        out.println("</div></body></html>");
    }
}
```

CookieServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.Cookie;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
@WebServlet("/CookieServlet")
public class CookieServlet extends HttpServlet {
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
        String name = "Guest";
        Cookie[] cookies = request.getCookies();
        if (cookies != null) {
            for (Cookie c : cookies) {
                if (c.getName().equals("userCookie")) {
                    name = c.getValue();
                }
            }
        }
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        out.println("<!DOCTYPE html><html><head><title>Cookie Result</title>");
        out.println("<style>body{font-family:Arial;margin:60px;background:#f4f6f8;}");
        out.println(".box{background:#fff;padding:25px 35px;border-radius:8px;}");
        out.println("box-shadow:0 0 10px rgba(0,0,0,0.15);max-width:400px;margin:auto;text-align:center;}");
        out.println("</style></head><body>");
        out.println("<div class='box'>");
        out.println("<h2>Hello, " + name + "</h2>");
        out.println("<p>(This value was retrieved from an HTTP Cookie stored in the browser.)</p>");
        out.println("</div></body></html>");
    }
}
```

Login

Username:

Password:

Login

Username or Password cannot be empty!

[Go Back](#)

Welcome, Vishwa

Total Unique Visitors So Far: 1

1. Hidden Form Field Demo

2. URL Rewriting Demo

[Visit \(URL Rewriting\)](#)

3. Cookie Demo

[Visit \(Cookies\)](#)

4. HttpSession Demo

[Visit \(HttpSession\)](#)

Hello, Vishwa

(This value was passed via a Hidden Form Field,
invisible to the user but submitted along with the form.)

Hello, Vishwa

(Passed via URL Rewriting query parameter.)

Hello, Vishwa

(This value was retrieved from an HTTP Cookie stored
in the browser.)

Hello, Vishwa

(This value was retrieved securely via server-side
HttpSession storage.)